Deep Feature-Based Text Clustering and Its Explanation

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Abstract

This work investigates unsupervised document clustering using deep pretrained text representations. In particular, document embeddings are extracted from pretrained encoders, and the impact of different pooling strategies (e.g., mean/max/last) and normalization schemes (e.g., identity, layer normalization, ℓ_2 normalization) on clustering quality is analyzed. The resulting representations are then grouped with a standard clustering procedure and evaluated using widely adopted clustering metrics (ACC, NMI, and ARI) on benchmark datasets of topical news and encyclopedic text. Beyond quantitative performance, the interpretability of the obtained partitions is studied through a post-hoc explanation approach based on a linear model, which identifies indicative terms for each cluster. Finally, low-dimensional visualizations (t-SNE) are reported to support a qualitative inspection of the separation among groups.

1 Introduction

Unsupervised text clustering aims to group documents by semantic similarity without relying on labeled data. However, performance strongly depends on the quality of the document representation and on modeling choices made along the pipeline. This work studies a feature-based clustering pipeline built on deep pretrained textual representations, focusing on the role of pooling and normalization strategies and on the interpretability of the resulting clusters. The experimental analysis is conducted on benchmark datasets and is complemented by post-hoc explanations based on indicative terms for each cluster.

2 Method

A deep feature-based text clustering pipeline is adopted, in which document representations are extracted from pretrained encoders, aggregated through pooling operations, optionally normalized, and subsequently clustered. In particular, contextualized embeddings obtained from ELMo and BERT are considered, and the impact of *pooling* and *normalization* choices on clustering quality is assessed under a consistent evaluation protocol. An overview of the entire framework is illustrated in Figure 1.

2.1 Representation Learning

Each document is encoded into a fixed-dimensional vector using a deep pretrained text encoder. We employ two widely used contextualized models, namely ELMo and BERT, to extract token-level embeddings.

Since encoders produce sequences of token representations, a document-level embedding is obtained by aggregating them through a pooling operation. We consider common strategies such as mean pooling, max pooling, and selecting a representative vector (e.g., the last hidden state). These choices influence how local lexical information and global semantics are captured, potentially affecting cluster compactness and separability.

After pooling, embeddings may be post-processed to modify the geometry of the representation space. We consider: (i) no normalization, (ii) feature-wise normalization (e.g., layer normalization), and (iii) vector normalization such as ℓ_2 normalization.

2.2 Clustering, Evaluation and Interpretability

Given the document embeddings, clustering is performed using two standard unsupervised algorithms: k-means and agglomerative clustering. Both methods rely on distance-based criteria in the embedding space, making them suitable for evaluating the impact of representation quality.

To ensure fair comparisons, the same clustering protocol is applied across all configurations. When ground-truth labels are available, they are used exclusively for evaluation. Performance is measured using clustering accuracy (ACC), Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI), with ACC computed after optimal label alignment.

To improve interpretability, a post-hoc explanation approach inspired by TCRE is adopted. A linear model is trained to predict cluster assignments from sparse lexical features (e.g., bag-of-words), and the learned weights are used to extract *indicative terms* for each cluster. These terms provide human-readable insights into cluster semantics and help identify failure cases.

For qualitative analysis, embeddings are visualized through two-dimensional t-SNE projections. As such projections may distort global distances, they are used to inspect local neighborhood structure rather than as quantitative evidence.

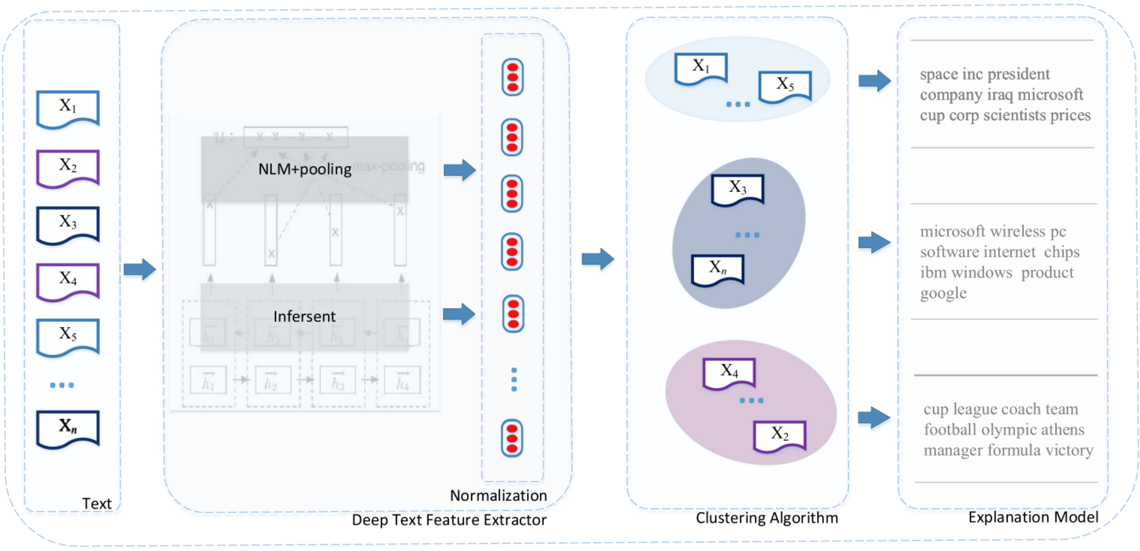

Figure 1: Overview of the proposed deep feature-based text clustering pipeline, including representation extraction, pooling, normalization, clustering, evaluation, and post-hoc explanation.

2.3 Experimental Setup

Datasets. Experiments are conducted on benchmark datasets of topical news and encyclopedic text (e.g., AG News, DBpedia, Yahoo Answers). The number of clusters is set equal to the number of ground-truth classes when available.

Preprocessing. Documents are minimally preprocessed to match the input requirements of the encoders. For post-hoc explanations, standard tokenization and vocabulary construction are applied consistently.

Experimental Factors. The study focuses on two representation-level choices: (i) pooling strategy (mean, max, last) and (ii) embedding normalization (identity, layer-wise, ℓ_2). All other components, including the encoder and clustering algorithm, are controlled to ensure a fair comparison.

Hyperparameters and Reproducibility. Experiments are conducted with a fixed random seed and, when applicable, averaged over multiple runs. Hyperparameters are selected using a consistent protocol.

Model Selection and Reporting. Configurations are compared using ACC, NMI, and ARI, with a consistent primary criterion (e.g., NMI or ARI). Results are reported in compact tables, while complete ablations are provided in the appendix. For interpretability, the top-k indicative terms per cluster are included for representative configurations.

3 ELMo + k-means Experiments

This section presents the experimental evaluation of document clustering based on ELMo embeddings combined with k-means. The analysis focuses on the impact of pooling strategies and normalization techniques on clustering performance.

Different configurations are obtained by varying: (i) the pooling strategy, including mean, max, and last token representations, and (ii) the embedding normalization, considering both no normalization and ℓ_2 normalization.

Clustering performance is evaluated using standard external metrics, namely clustering accuracy (ACC), Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI). When available, results are compared with those reported in the reference paper in order to assess consistency.

3.1 Overall Results

Table 1 reports the performance obtained across the considered configurations, alongside the corresponding values from the reference paper.

The results highlight the impact of representation-level design choices on clustering quality. In particular, both pooling strategy and normalization affect the structure of the embedding space and, consequently, the separability of clusters.

In the following, each configuration is analyzed in detail. For every experiment, t-SNE visualizations are provided to qualitatively assess the embedding space, together with a discussion of clustering behavior and a comparison with the corresponding results reported in the literature.

3.2 ELMo + Mean Pooling + Layer Normalization

This experiment evaluates document clustering on the AG News dataset using ELMo embeddings with mean pooling, followed by layer normalization, and k-means clustering. A balanced subset of the dataset is considered, with an equal number of samples per class.

The obtained clustering performance is as follows:

Dataset	Method	ACC	NMI	ARI
AG News	ELMo + Mean + LN (KMeans)	**83.08**	**58.70**	**61.37**
	ELMo + Mean + LN (Agglo)	81.67	55.86	57.91
	ELMo + Mean + L2 (KMeans)	82.97	58.59	61.21
	ELMo + Mean + L2 (Agglo)	81.23	55.86	58.00
	BERT + Max + LN (KMeans)	80.45	54.46	56.34
	BERT + Max + L2 (KMeans)	61.00	55.33	57.18
DBpedia	ELMo + Mean + I (KMeans)	74.34	75.39	65.56
	ELMo + Mean + I (Agglo)	64.23	71.89	56.17
	ELMo + Mean + L2 (KMeans)	**74.61**	**75.85**	**64.96**
	ELMo + Mean + L2 (Agglo)	69.01	73.69	61.21
	BERT + Max + LN (KMeans)	67.99	73.65	53.95
	BERT + Max + I (KMeans)	34.53	44.05	20.95
	BERT + Max + L2 (KMeans)	64.89	67.16	46.21
Yahoo	ELMo + Mean + L2 (KMeans)	**44.74**	**30.77**	**21.76**
	ELMo + Mean + LN (Agglo)	45.48	31.34	22.43
	BERT + Max + L2 (KMeans)	20.21	7.90	3.19
R5	ELMo + Mean + LN (KMeans)	**85.06**	**69.38**	**71.51**
	ELMo + Mean + LN (Agglo)	80.74	67.07	67.98
	BERT + Max + L2 (KMeans)	63.51	48.81	48.37
Emotion	ELMo + Mean + LN (KMeans)	**21.91**	**1.97**	**1.11**
	ELMo + Mean + LN (Agglo)	21.03	1.51	0.61
	BERT + Max + L2 (KMeans)	19.94	0.61	0.33

Table 1: Clustering performance per dataset, best results are highlighted in bold.

- **ACC**: 83.08%
- **NMI**: 58.70%
- **ARI**: 61.37%

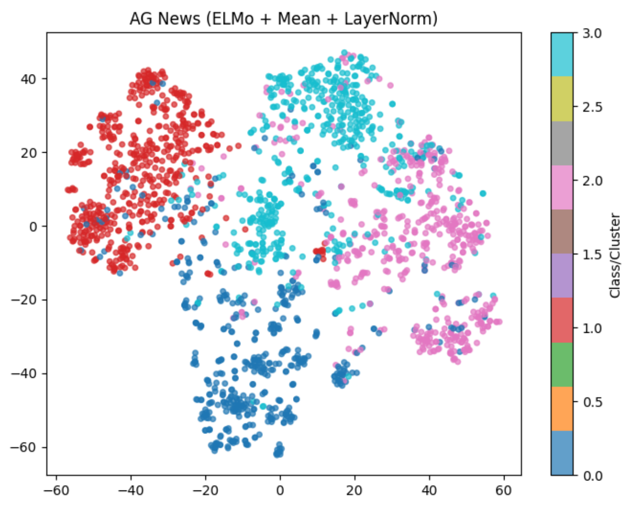

Figure 2: t-SNE visualization of ELMo embeddings with mean pooling and layer normalization on AG News.

The t-SNE visualization reveals a partially well-structured embedding space. One cluster (left side of the plot) appears clearly separated from the others, indicating that the corresponding class is well captured by the representation. Another cluster forms a relatively compact group in the lower region, suggesting a good level of semantic coherence.

In contrast, the remaining clusters exhibit substantial overlap, particularly in the central-right area of the projection. This behavior suggests that the corresponding categories share similar lexical and semantic features, making them harder to distinguish. In the context of AG News, this effect is typically observed between categories such as Business and Sci/Tech, which often rely on overlapping terminology.

Overall, the results indicate that mean pooling combined with layer normalization provides a reasonably effective representation, enabling clear separation for some classes while still showing limitations in distinguishing semantically related categories.

3.3 ELMo + Mean Pooling + No Normalization

This experiment evaluates document clustering on the DBpedia dataset using ELMo embeddings with mean pooling and no additional normalization, followed by k-means clustering. A balanced subset of the dataset is considered, with an equal number of samples per class.

The obtained clustering performance is as follows:

- **ACC**: 74.34%
- **NMI**: 75.39%
- **ARI**: 65.56%

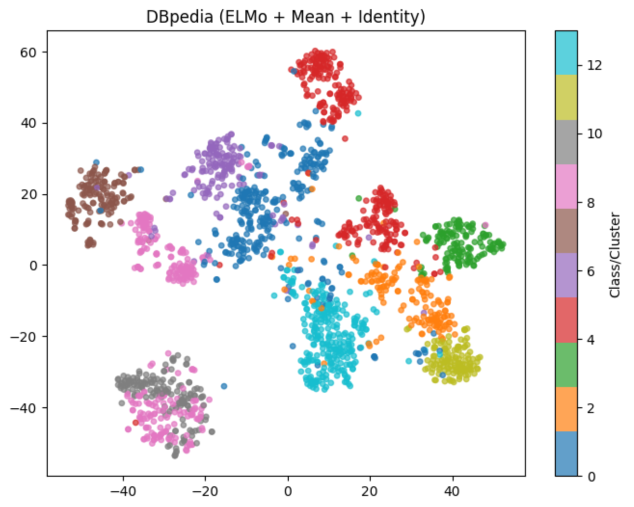

Figure 3: t-SNE visualization of ELMo embeddings with mean pooling and no normalization on DBpedia.

The t-SNE visualization reveals a heterogeneous clustering structure. Several clusters appear well separated in the embedding space, with clearly isolated groups located in different regions of the projection. This indicates that ELMo is capable of producing highly distinctive representations for certain categories.

At the same time, some classes exhibit fragmented structures, appearing as multiple sub-clusters or scattered regions. This behavior is typically associated with semantically broad categories, whose instances span diverse topics and contexts.

Moreover, regions of significant overlap are observed, particularly in the lower-left area of the plot. This suggests that, for some classes, mean pooling alone is not sufficient to ensure clear separability. The absence of normalization may contribute to this effect, as variations in embedding magnitude can distort the geometry of the feature space and reduce the effectiveness of distance-based clustering.

Overall, the results show that, while mean pooling without normalization can still produce reasonably structured embeddings, it leads to less consistent cluster separation compared to configurations that include normalization.

3.4 ELMo + Mean Pooling + ℓ_2 Normalization

This experiment evaluates document clustering using ELMo embeddings with mean pooling and ℓ_2 normalization, followed by k-means clustering. The analysis is conducted on both AG News and DBpedia datasets, using balanced subsets with a fixed number of samples per class.

The obtained clustering performance is as follows:

AG News

- **ACC**: 82.97%
- **NMI**: 58.59%
- **ARI**: 61.21%

DBpedia

- **ACC**: 74.61%
- **NMI**: 75.85%
- **ARI**: 64.96%

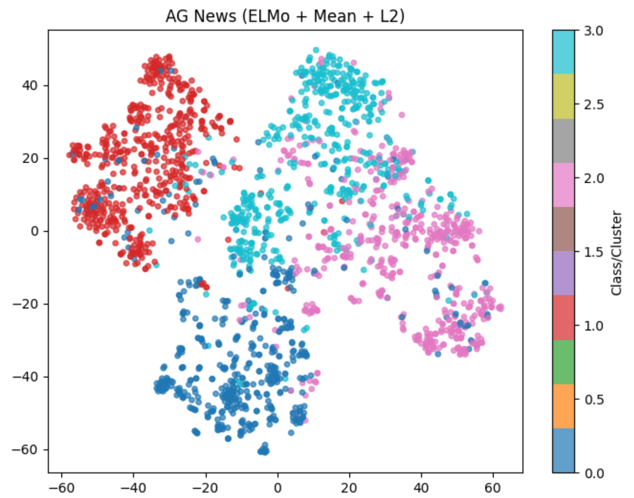

Figure 4: t-SNE visualization of ELMo embeddings with mean pooling and ℓ_2 normalization on AG News.

The t-SNE visualization for AG News shows a less cohesive structure compared to the LayerNorm configuration. The clusters appear more fragmented, with several small sub-groups distributed around the main regions. This suggests that ℓ_2 normalization does not fully preserve the global structure of the embedding space, leading to less compact clusters.

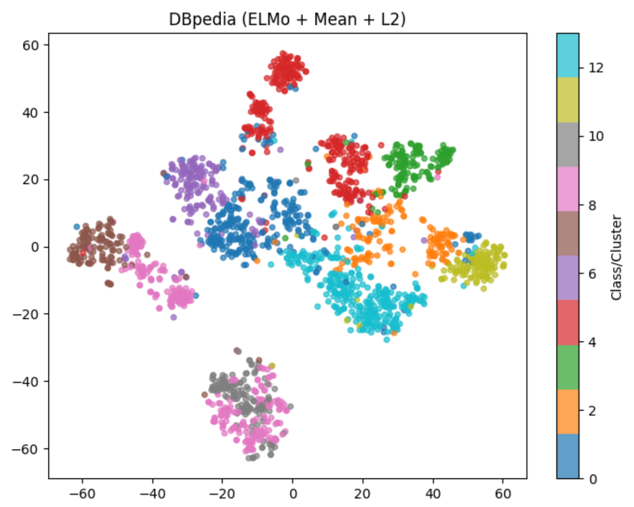

Figure 5: t-SNE visualization of ELMo embeddings with mean pooling and ℓ_2 normalization on DBpedia.

In contrast, the DBpedia visualization shows slightly more compact clusters compared to the identity normalization setting. The ℓ_2 normalization reduces the influence of outliers by constraining all vectors to lie on the unit sphere, effectively pulling isolated points closer to the central mass of their respective clusters.

Overall, the results indicate that ℓ_2 normalization has a dataset-dependent effect. While it may introduce fragmentation in simpler datasets such as AG News, it can improve compactness and reduce dispersion in more complex datasets such as DBpedia.

3.5 ELMo + Mean Pooling + ℓ_2 Normalization on Yahoo

This experiment evaluates document clustering on the Yahoo Answers dataset using ELMo embeddings with mean pooling and ℓ_2 normalization, followed by k-means clustering. A balanced subset of the dataset is considered, with a fixed number of samples per class.

The obtained clustering performance is as follows:

- **ACC**: 44.74%
- **NMI**: 30.77%
- **ARI**: 21.76%

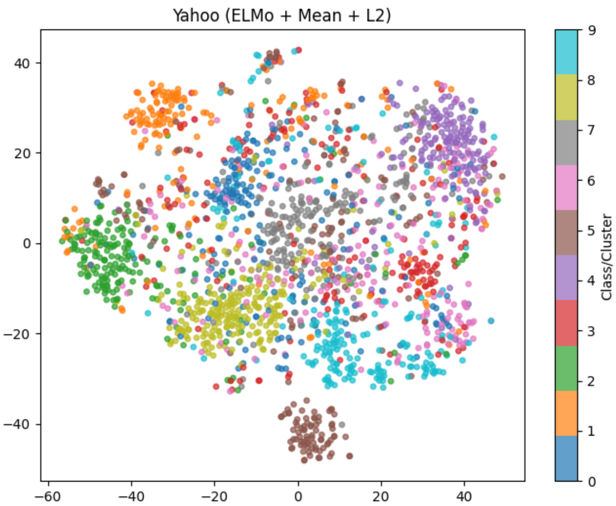

Figure 6: t-SNE visualization of ELMo embeddings with mean pooling and ℓ_2 normalization on Yahoo Answers.

The t-SNE visualization reveals a markedly different structure compared to previous datasets. Instead of well-separated clusters, the embedding space is characterized by a dense central region where data points from different classes are heavily intermixed. This "nebula-like" structure indicates a lack of clear separability between categories.

Only a few classes exhibit partial separation from the main mass, forming loosely defined peripheral groups. However, even in these cases, a significant number of points from other classes are interspersed within the clusters, reducing their coherence.

This behavior can be attributed to the high semantic overlap of the Yahoo Answers dataset. Categories such as *Family & Relationships*, *Society & Culture*, and *Politics & Government* share similar vocabulary and informal language patterns, making them difficult to distinguish using mean-pooled ELMo embeddings.

Overall, the results highlight a limitation of the considered representation. In the presence of semantically overlapping and noisy categories, mean pooling combined with ℓ_2 normalization is insufficient to produce well-separated clusters, leading to a significant drop in clustering performance.

3.6 ELMo + Mean Pooling + Layer Normalization on R5

This experiment evaluates document clustering on the Reuters R5 subset using ELMo embeddings with mean pooling and layer normalization, followed by k-means clustering. A balanced subset of the dataset is considered, with a fixed number of samples per class.

The obtained clustering performance is as follows:

- **ACC**: 85.06%
- **NMI**: 69.38%
- **ARI**: 71.51%

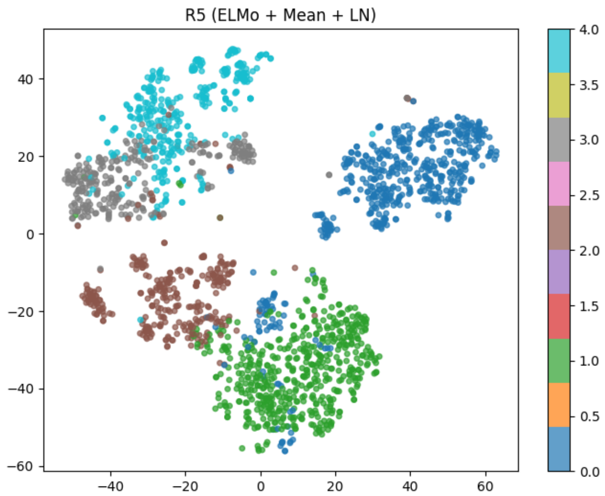

Figure 7: t-SNE visualization of ELMo embeddings with mean pooling and layer normalization on Reuters R5.

The t-SNE visualization shows a clearly structured embedding space, where most classes form well-defined and spatially separated clusters. Several "island-like" regions can be observed, with minimal overlap between different categories.

Compared to more semantically heterogeneous datasets such as Yahoo Answers, the Reuters R5 subset exhibits stronger cluster separability. This behavior can be attributed to the more focused and domain-specific nature of newswire text, where categories are characterized by distinct lexical patterns (e.g., economic or corporate terminology).

Layer normalization appears to be particularly effective in this setting, as it promotes compactness within clusters while preserving inter-class distances. This results in embedding geometries that are highly suitable for distance-based clustering methods such as k-means.

Overall, this configuration achieves the most stable and well-separated clustering structure among the considered datasets, both in terms of quantitative metrics and qualitative visualization.

4 Experiments with BERT

The evaluation is extended to a Transformer-based pretrained encoder, **BERT (bert-base-uncased)**, used as a feature extractor for document clustering. BERT is a bidirectional Transformer model that produces context-aware token representations, capable of capturing complex semantic and syntactic dependencies. Due to its strong performance across a wide range of NLP tasks, BERT represents a natural candidate for evaluating deep feature-based clustering approaches.

Document embeddings are extracted from the last hidden states of the model and aggregated via max pooling over the token dimension. The resulting representations are then normalized using layer normalization prior to clustering. Clustering is performed using k-means with $n_{\text{init}} = 20$ to ensure stable centroid initialization.

Experiments are conducted on the AG News and DBpedia datasets. Clustering performance is evaluated using the same metrics introduced in previous sections, namely Clustering Accuracy (ACC), Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI), allowing for a consistent comparison with ELMo-based configurations.

4.1 BERT + Max Pooling + Layer Normalization on AG News

This experiment evaluates document clustering on the AG News dataset using BERT (bert-base-uncased) as a feature extractor. Token representations are obtained from the last hidden states and aggregated via max pooling. The resulting embeddings are normalized using layer normalization before applying k-means clustering.

The obtained clustering performance is as follows:

- **ACC**: 80.45%
- **NMI**: 54.46%
- **ARI**: 56.34%

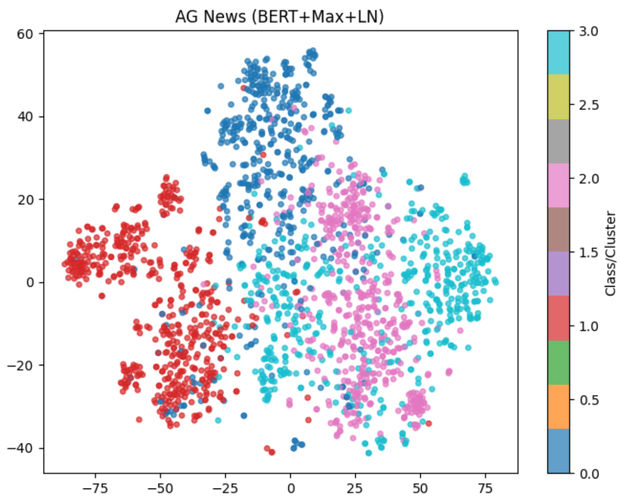

Figure 8: t-SNE visualization of BERT embeddings with max pooling and layer normalization on AG News.

The t-SNE visualization reveals a cluster geometry that differs from that obtained with ELMo-based representations. In particular, clusters appear elongated and anisotropic, forming filament-like structures rather than compact, spherical regions. This shape is challenging for k-means, which assumes isotropic clusters, and partially explains the observed performance.

A central region of the embedding space shows significant overlap between multiple classes, especially between semantically related categories such as Business and Sci/Tech. This indicates that, despite the contextual strength of BERT representations, fine-grained distinctions between these domains remain difficult in AG News.

In contrast, one class appears consistently well separated from the others, suggesting that BERT is able to capture highly discriminative features for certain topics (e.g., World or Sports), similarly to what was observed with ELMo-based embeddings.

Overall, the results suggest that BERT produces richer but more geometrically complex embedding spaces, where cluster separability is influenced not only by semantic content but also by structural anisotropy in the representation space.

4.2 BERT + Max Pooling + Layer Normalization on DBpedia

This experiment evaluates BERT-based document clustering on the DBpedia dataset. Document embeddings are extracted from the last hidden states of the model and aggregated using max pooling over the token dimension. The resulting representations are normalized using layer normalization prior to applying k-means clustering.

The clustering performance obtained is reported below:

- **ACC**: 67.99%
- **NMI**: 73.65%
- **ARI**: 53.95%

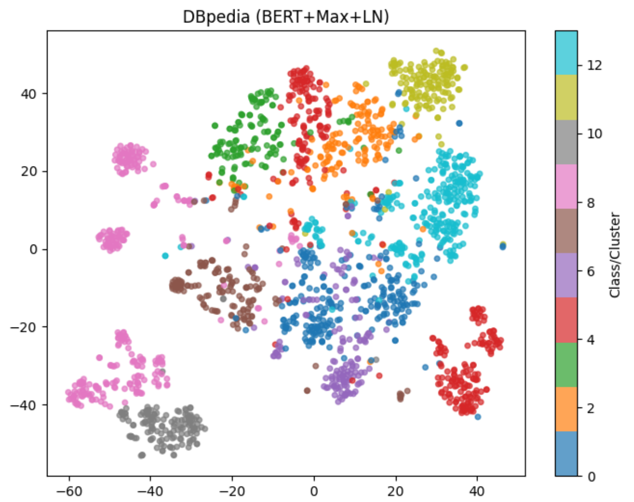

Figure 9: t-SNE visualization of BERT embeddings with max pooling and layer normalization on DBpedia.

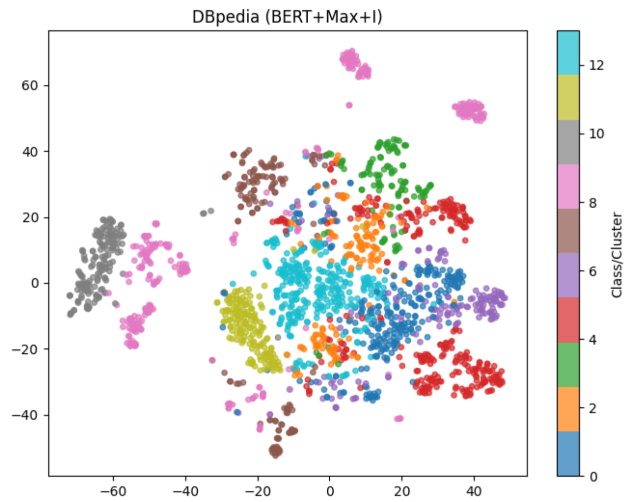

Figure 10: t-SNE visualization of BERT embeddings with max pooling and no normalization on DBpedia.

The visualization highlights a more complex embedding geometry compared to AG News. Several classes appear fragmented into multiple sub-clusters, suggesting that BERT captures fine-grained lexical and semantic variations within the same high-level category. This is particularly evident in broad classes such as entities related to locations or creative works.

A dense central region emerges where multiple classes overlap significantly, indicating that semantic boundaries are less well-defined in this region of the embedding space. This overlap negatively impacts centroid-based clustering methods such as k-means, which rely on convex cluster assumptions.

Despite this, some classes remain clearly separated in peripheral regions of the space, suggesting that BERT still encodes highly discriminative features for specific categories (e.g., animals or species-related entities). Overall, DBpedia reveals a trade-off between representational richness and clustering separability when using BERT embeddings.

4.3 BERT + Max Pooling + No Normalization on DBpedia

This experiment evaluates BERT-based document clustering on the DBpedia dataset without applying any post-processing normalization. Document embeddings are obtained from the last hidden states and aggregated via max pooling, followed directly by k-means clustering.

The clustering performance is significantly lower compared to the normalized setting:

- **ACC**: 34.53%
- **NMI**: 44.05%
- **ARI**: 20.95%

The t-SNE visualization reveals a highly disorganized embedding space. Many classes are fragmented into multiple disconnected regions, often appearing as small sub-clusters scattered across distant areas of the projection. This indicates a lack of global coherence in the representation.

A dense central region is also observed, where points from several classes are heavily intermixed. From the perspective of k-means, this area effectively collapses into a single indistinguishable cluster, severely limiting clustering performance.

Compared to the layer-normalized setting, the embedding space appears significantly more dispersed, with points spread unevenly across the projection. This behavior highlights the importance of normalization in shaping the geometry of BERT representations. Without it, variations in vector magnitude and direction lead to poorly structured feature spaces that are not well suited for distance-based clustering.

Overall, the results demonstrate that normalization is a critical component when using BERT embeddings for clustering, as it helps enforce a more regular and separable structure in the representation space.

4.4 BERT + Max Pooling + ℓ_2 Normalization

This experiment evaluates BERT-based document clustering using max pooling and ℓ_2 normalization, followed by k-means clustering. The analysis is conducted on both AG News and DBpedia datasets.

The obtained clustering performance is reported below:

AG News

- **ACC**: 81.00%
- **NMI**: 55.33%
- **ARI**: 57.18%

DBpedia

- **ACC**: 64.89%
- **NMI**: 67.16%
- **ARI**: 46.21%

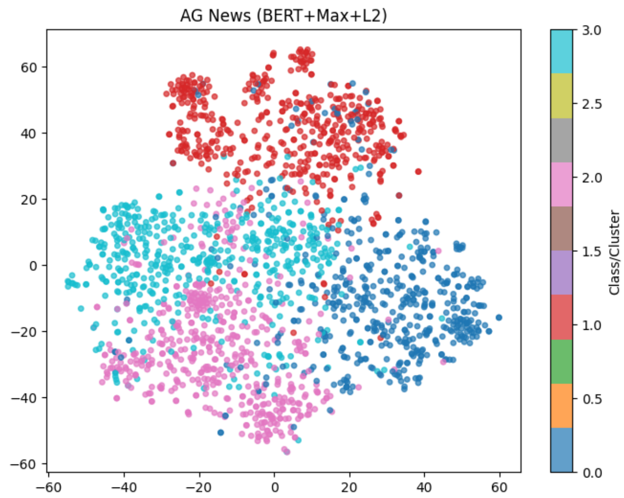

Figure 11: t-SNE visualization of BERT embeddings with max pooling and ℓ_2 normalization on AG News.

For AG News, the embedding space exhibits moderately compact clusters. One class remains clearly separated from the others, while semantically related categories (such as Business and Sci/Tech) continue to overlap in the central region. Compared to the layer-normalized configuration, ℓ_2 normalization produces slightly denser clusters but does not fully resolve inter-class ambiguity.

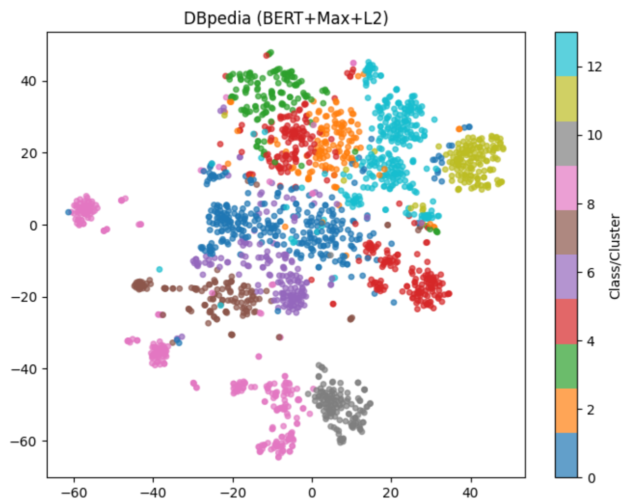

Figure 12: t-SNE visualization of BERT embeddings with max pooling and ℓ_2 normalization on DBpedia.

On DBpedia, the visualization reveals increased dispersion of data points across the embedding space. Several classes appear fragmented into multiple regions, which complicates the identification of well-defined centroids. This behavior suggests that, while ℓ_2 normalization constrains vector magnitudes, it does not fully address the intrinsic complexity of BERT representations in high-diversity datasets.

Overall, ℓ_2 normalization provides a partial improvement over the non-normalized setting, but remains less effective than layer normalization in producing stable and well-structured clusters.

4.5 BERT + Max Pooling + ℓ_2 Normalization on Yahoo

This experiment evaluates BERT-based document clustering on the Yahoo Answers dataset using max pooling and ℓ_2 normalization, followed by k-means clustering.

The obtained clustering performance is significantly lower compared to previous datasets:

- **ACC**: 20.21%
- **NMI**: 7.90%
- **ARI**: 3.19%

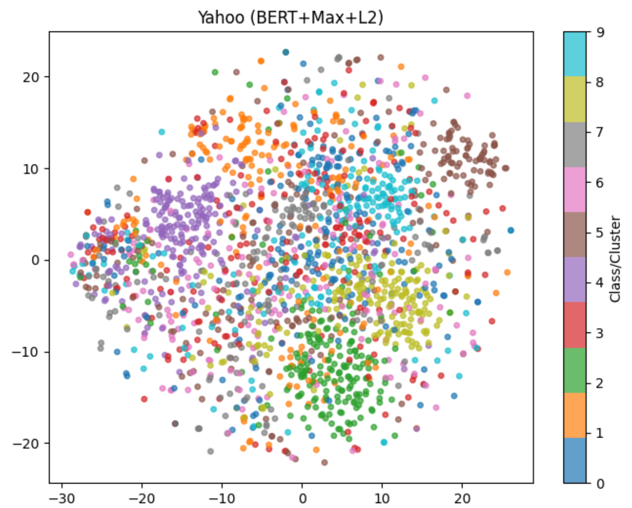

Figure 13: t-SNE visualization of BERT embeddings with max pooling and ℓ_2 normalization on Yahoo Answers.

The t-SNE visualization reveals a complete lack of cluster structure. Data points form a dense, uniform cloud with no clearly identifiable regions, and samples from different classes are extensively intermixed across the entire space.

This behavior can be attributed to a combination of factors. First, the Yahoo Answers dataset exhibits high semantic overlap across categories, with informal and heterogeneous language that reduces discriminability. Second, BERT embeddings are known to exhibit anisotropy, meaning that representations tend to occupy a narrow region of the vector space. When combined with max pooling over long and diverse texts, this effect is amplified, leading to highly similar document representations.

The application of ℓ_2 normalization does not alleviate this issue, as it only constrains vector magnitudes without improving directional separability. As a result, the embedding space remains poorly structured, making it unsuitable for centroid-based clustering methods such as k-means.

Overall, this experiment highlights a failure case for BERT-based clustering, where both the nature of the dataset and the geometry of the representation space prevent the emergence of meaningful clusters.

4.6 BERT + Max Pooling + ℓ_2 Normalization on R5

This experiment evaluates BERT-based document clustering on the Reuters R5 dataset using max pooling and ℓ_2 normalization, followed by k-means clustering.

The obtained clustering performance is as follows:

- **ACC**: 63.51%
- **NMI**: 48.81%
- **ARI**: 48.37%

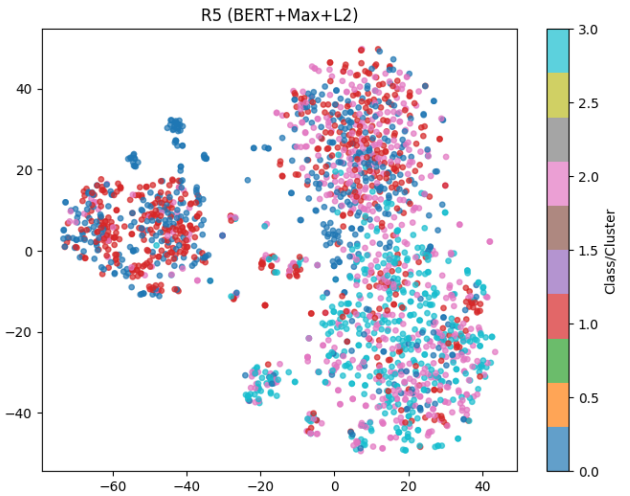

Figure 14: t-SNE visualization of BERT embeddings with max pooling and ℓ_2 normalization on Reuters R5.

The t-SNE visualization reveals a partially structured but highly overlapping embedding space. Instead of clearly separated clusters, the data points form a few large regions where multiple classes are intermixed, reducing overall separability.

Compared to the ELMo-based configuration on the same dataset, the cluster structure appears less defined. While some degree of grouping is visible, class boundaries are not clearly delineated, leading to confusion between semantically related categories.

This behavior can be attributed to the interaction between BERT representations and max pooling. The pooling operation may emphasize tokens that are locally salient but not globally discriminative across classes, especially in domains characterized by shared vocabulary such as financial news. As a result, different categories (e.g., *earn*, *acq*, *crude*) may be mapped to similar regions of the embedding space.

Additionally, fragmentation effects can be observed, where subsets of the same class form isolated micro-clusters. This further complicates centroid-based clustering, as k-means is forced to partition irregular and overlapping structures.

Overall, the results suggest that, despite the expressive power of BERT, the combination of max pooling and ℓ_2 normalization does not yield well-separated clusters on this dataset, highlighting the importance of representation geometry and pooling choices in clustering performance.

5 Experiments with Agglomerative Clustering

This section evaluates the robustness of the proposed pipeline with respect to the choice of clustering algorithm. To this end, the k-means procedure used in previous experiments is replaced with an alternative method, while keeping the feature extraction pipeline unchanged.

In particular, agglomerative clustering is considered as a hierarchical baseline. Unlike k-means, which assumes spherical clusters and relies on centroid-based partitioning, agglomerative methods iteratively merge samples based on pairwise similarity, allowing for more flexible cluster shapes in the embedding space.

Clustering is performed on the same document representations used in previous sections, with identical pooling and normalization strategies. The number of clusters is fixed to the number of ground-truth classes to ensure consistency across experiments.

Performance is evaluated using clustering accuracy (ACC), Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI), enabling a direct comparison with k-means-based results.

5.1 ELMo + Mean Pooling + Layer Normalization with Agglomerative Clustering on AG News

This experiment evaluates document clustering on the AG News dataset using ELMo embeddings with mean pooling and layer normalization, replacing k-means with agglomerative clustering.

The obtained clustering performance is as follows:

- **ACC**: 81.67%
- **NMI**: 55.86%
- **ARI**: 57.91%

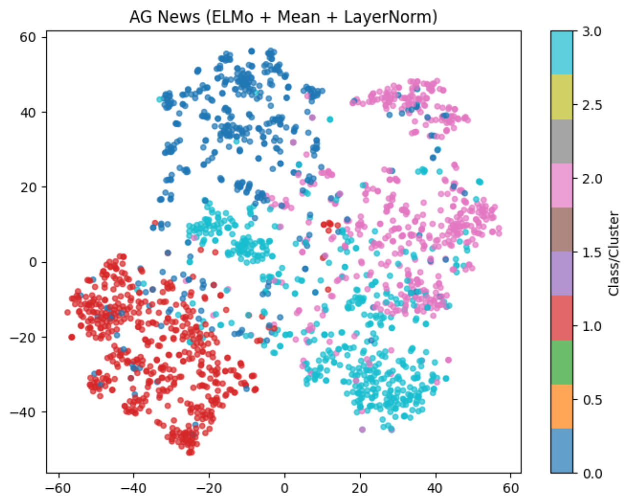

Figure 15: t-SNE visualization of ELMo embeddings with mean pooling and layer normalization on AG News.

The t-SNE visualization confirms the overall structure observed in previous experiments. Two classes remain clearly separated from the others, forming well-defined regions in the embedding space. These clusters are characterized by distinctive lexical patterns, making them easier to identify regardless of the clustering method.

In contrast, a substantial overlap persists between two central classes, which appear heavily intertwined. This overlap reflects strong semantic similarity and remains a critical challenge for clustering algorithms.

Compared to k-means, agglomerative clustering operates by iteratively merging samples based on pairwise distances. In this setting, such behavior may lead to early merging of semantically similar but distinct categories, particularly in overlapping regions. This can result in less precise cluster boundaries and slightly reduced performance.

Overall, the results indicate that, for this configuration, agglomerative clustering achieves comparable but slightly lower performance than k-means, suggesting that the underlying representation remains the dominant factor in determining clustering quality.

5.2 ELMo + Mean Pooling + No Normalization with Agglomerative Clustering on DBpedia

This experiment evaluates document clustering on the DBpedia dataset using ELMo embeddings with mean pooling and no normalization, combined with agglomerative clustering.

The obtained clustering performance is as follows:

- **ACC**: 64.23%
- **NMI**: 71.89%
- **ARI**: 56.17%

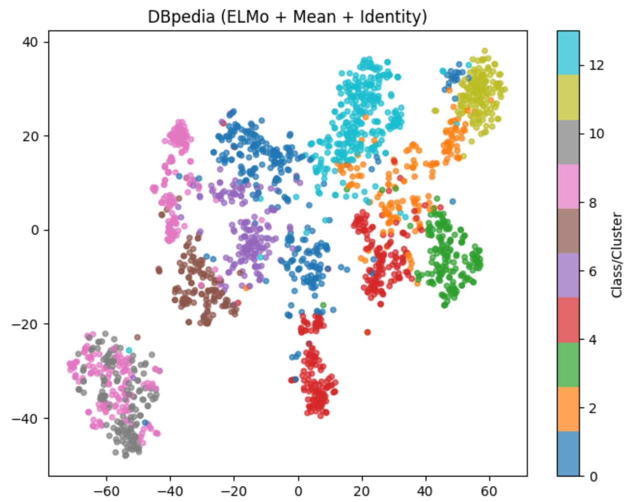

Figure 16: t-SNE visualization of ELMo embeddings with mean pooling and no normalization on DBpedia.

The t-SNE visualization highlights the complexity of clustering a dataset with a large number of classes. Several categories appear fragmented into multiple disconnected regions, indicating that samples belonging to the same class may occupy distant areas in the embedding space.

In addition, regions of high density and overlap are observed, particularly in central and lower areas of the projection. In these zones, multiple classes are closely intertwined, resulting in blurred cluster boundaries.

Compared to k-means, agglomerative clustering struggles to reconcile fragmented structures. Since the algorithm relies on local pairwise distances, it may fail to correctly merge distant sub-clusters belonging to the same class. At the same time, in dense regions, early merging decisions can group together samples from different classes, reducing overall clustering accuracy.

Overall, the results suggest that agglomerative clustering is more sensitive to fragmentation and local density variations in the embedding space, leading to lower performance compared to centroid-based methods in this setting.

5.3 ELMo + Mean Pooling + ℓ_2 Normalization with Agglomerative Clustering

This experiment evaluates ELMo-based document clustering with mean pooling and ℓ_2 normalization, combined with agglomerative clustering. The analysis is conducted on both AG News and DBpedia datasets.

The obtained clustering performance is reported below:

AG News

- **ACC**: 81.23%
- **NMI**: 55.86%
- **ARI**: 58.00%

DBpedia

- **ACC**: 69.01%
- **NMI**: 73.69%
- **ARI**: 61.21%

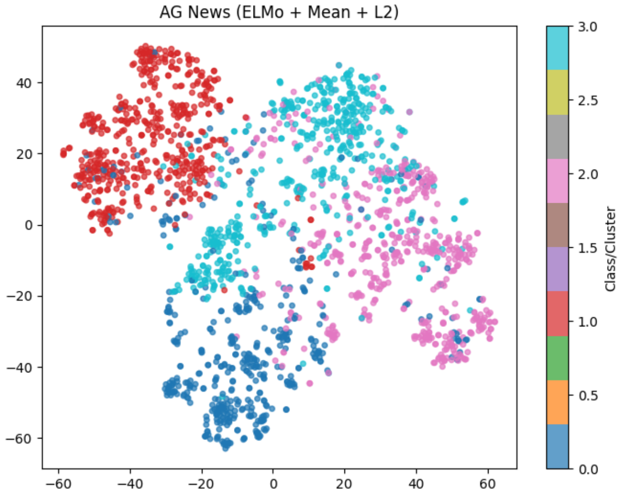

Figure 17: t-SNE visualization of ELMo embeddings with mean pooling and ℓ_2 normalization on AG News.

For AG News, the overall cluster structure remains consistent with previous configurations. Two classes continue to form well-separated regions, while the central area exhibits overlap between semantically related categories. The use of ℓ_2 normalization leads to slightly more compact clusters, but does not significantly alter the separability of the embedding space.

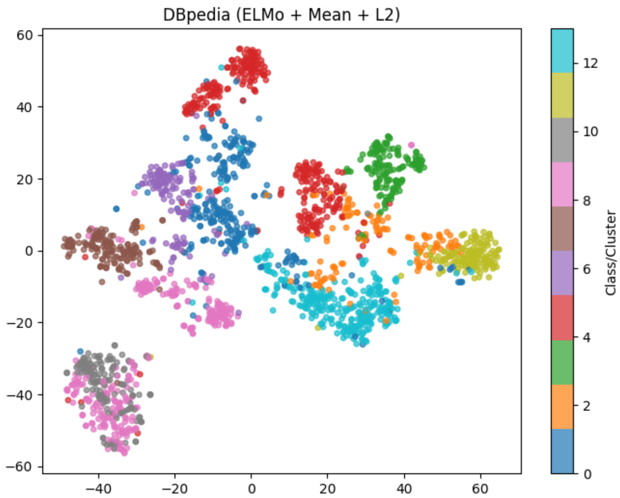

Figure 18: t-SNE visualization of ELMo embeddings with mean pooling and ℓ_2 normalization on DBpedia.

On DBpedia, the effect of ℓ_2 normalization is more pronounced. The embedding space exhibits a clearer organization into well-defined regions, with several classes forming compact and spatially separated clusters. This reduction in dispersion helps mitigate fragmentation effects observed in previous settings.

Compared to the identity normalization case, agglomerative clustering benefits from the more regular geometry induced by ℓ_2 normalization. By reducing the influence of vector magnitude and emphasizing directional similarity, the algorithm is able to produce more coherent clusters.

Overall, the results suggest that ℓ_2 normalization improves the compatibility between ELMo representations and agglomerative clustering, particularly in datasets with a large number of classes such as DBpedia.

5.4 ELMo + Mean Pooling + ℓ_2 Normalization with Agglomerative Clustering on Yahoo

This experiment evaluates document clustering on the Yahoo Answers dataset using ELMo embeddings with mean pooling and ℓ_2 normalization, combined with agglomerative clustering.

The obtained clustering performance is as follows:

- **ACC**: 45.48%
- **NMI**: 31.34%
- **ARI**: 22.43%

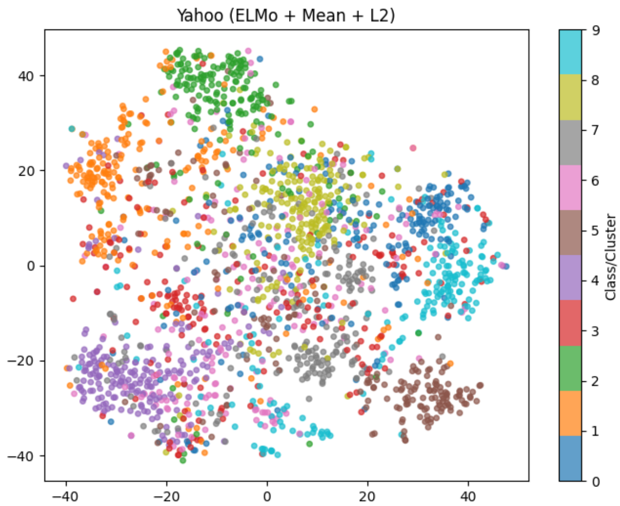

Figure 19: t-SNE visualization of ELMo embeddings with mean pooling and ℓ_2 normalization on Yahoo Answers.

The t-SNE visualization reveals a partially structured embedding space. Several regions exhibit compact groups of points with a dominant class, suggesting the presence of localized semantic coherence. These clusters appear more defined compared to previous experiments on the same dataset.

However, a large central region remains highly mixed, with significant overlap between multiple classes. This reflects the intrinsic ambiguity of the Yahoo Answers dataset, where categories share vocabulary and exhibit strong semantic overlap.

Compared to k-means, agglomerative clustering produces a more granular partitioning of the space, capturing local groupings more effectively. This is consistent with its hierarchical nature, which allows the identification of smaller coherent structures within a globally ambiguous space.

Overall, while clustering performance remains limited due to dataset complexity, the results indicate that agglomerative clustering can better exploit local structure in challenging scenarios, even when global cluster separation is weak.

5.5 ELMo + Mean Pooling + Layer Normalization with Agglomerative Clustering on R5

This experiment evaluates document clustering on the Reuters R5 dataset using ELMo embeddings with mean pooling and layer normalization, combined with agglomerative clustering.

The obtained clustering performance is as follows:

- **ACC**: 80.74%
- **NMI**: 67.07%
- **ARI**: 67.98%

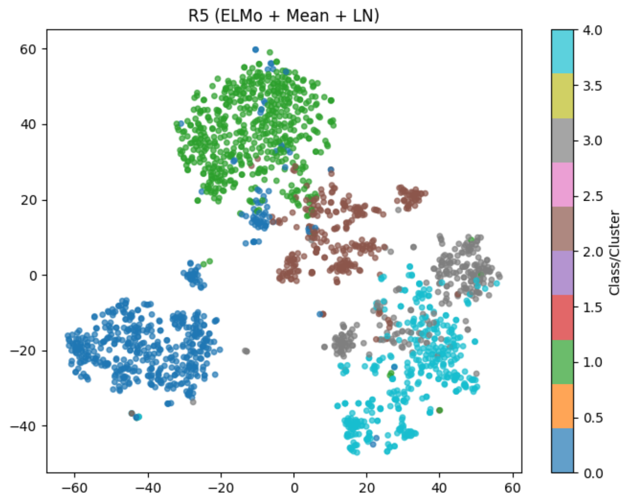

Figure 20: t-SNE visualization of ELMo embeddings with mean pooling and layer normalization on Reuters R5.

The t-SNE visualization shows a highly structured embedding space, where the five classes form clearly separated regions. Each class occupies a compact and well-defined area, with minimal overlap between clusters.

The separation between clusters is particularly evident for the most distant classes, which appear as almost completely isolated "islands" in the embedding space. This confirms the suitability of the Reuters R5 dataset for evaluating clustering methods, due to its strong topical coherence.

Some limited interaction between clusters is observed in boundary regions, where classes come into close proximity. In these areas, agglomerative clustering may be sensitive to local bridges of points, which can influence the hierarchical merging process and lead to occasional misassignments.

Overall, the results indicate that when the embedding space is well-structured, agglomerative clustering is capable of achieving strong performance, closely matching centroid-based methods while benefiting from its ability to capture hierarchical relationships.

6 Experiments on Emotion Dataset (Short Text Clustering)

Clustering is further evaluated on the Emotion dataset, a collection of short text sequences annotated with six fine-grained emotion categories. This setting is particularly challenging due to the limited context available in each sample and the high semantic ambiguity typical of short texts.

The dataset is used to assess the quality of learned representations in low-context scenarios, where subtle lexical and syntactic cues must be captured to discriminate between closely related emotional states.

Document representations are extracted using BERT (bert-base-uncased) with max pooling applied over token embeddings, followed by layer normalization. This configuration is chosen to emphasize the most salient token-level signals while ensuring a stable embedding geometry for distance-based clustering.

To evaluate the robustness of the representations, both k-means and agglomerative clustering are applied. Performance is measured using clustering accuracy (ACC), Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI), enabling a consistent comparison across clustering strategies in a short-text regime.

This experimental setting allows for the analysis of how well pretrained language model embeddings capture fine-grained emotional semantics under strong contextual constraints.

6.1 ELMo + Mean Pooling + Layer Normalization with K-Means on Emotion Dataset

This experiment evaluates clustering performance on the Emotion dataset using ELMo-based embeddings with mean pooling and layer normalization, followed by k-means clustering with six clusters corresponding to the ground-truth emotion categories.

The obtained results are the following:

- **ACC**: 21.91%
- **NMI**: 1.97%
- **ARI**: 1.11%

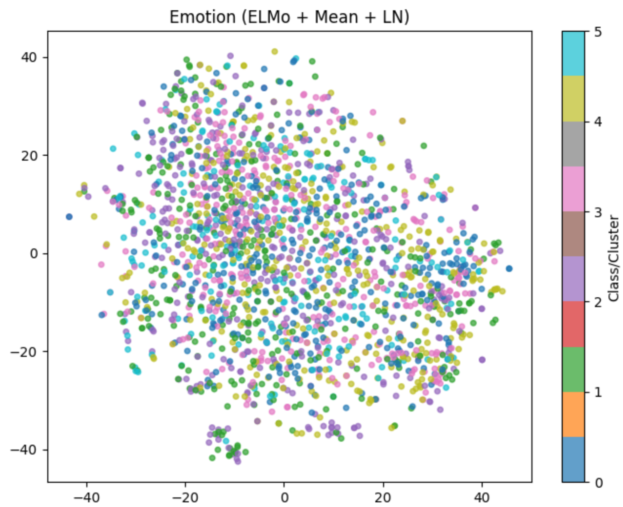

Figure 21: t-SNE visualization of ELMo embeddings with mean pooling and layer normalization on the Emotion dataset.

The t-SNE visualization reveals a highly entangled embedding space, where no clearly separable structure emerges across the six emotion categories. The points form a dense and homogeneous cloud, with extensive overlap between all classes.

Unlike previous datasets, no meaningful "island-like" structures are observed, indicating that the learned representation does not naturally partition emotional categories in the embedding space.

This behavior suggests that, in short-text settings with subtle semantic differences, ELMo embeddings combined with mean pooling and layer normalization fail to produce sufficiently discriminative representations for clustering. As a result, k-means is unable to identify meaningful centroids, leading to near-random assignment performance.

6.2 ELMo + Mean Pooling + Layer Normalization with Agglomerative Clustering on Emotion Dataset

This experiment evaluates the performance of agglomerative clustering on the Emotion dataset using ELMo-based embeddings with mean pooling and layer normalization. The number of clusters is set to six, corresponding to the ground-truth emotion categories.

The obtained results are as follows:

- **ACC**: 21.03%
- **NMI**: 1.51%
- **ARI**: 0.61%

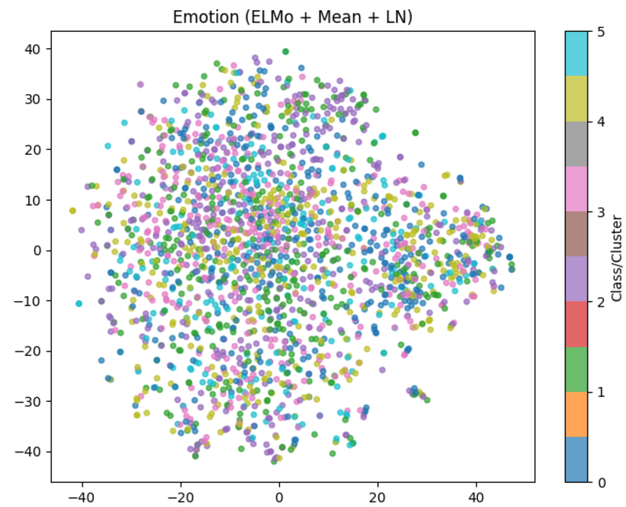

Figure 22: t-SNE visualization of ELMo embeddings with mean pooling and layer normalization on the Emotion dataset (agglomerative clustering setting).

The t-SNE visualization confirms a fully entangled embedding space, where no meaningful local structure or class-specific regions can be identified. The distribution appears approximately isotropic, with extensive overlap among all six emotion categories.

Unlike previous datasets, no dense class-specific regions emerge, and local neighborhoods contain a uniform mixture of labels. This indicates that the embedding space does not preserve emotion-related semantic distances.

As a result, agglomerative clustering fails to recover any meaningful hierarchical structure, since no stable local groupings exist to support the merging process. The extremely low NMI and ARI values further confirm that the clustering performance is close to random assignment in this setting.

6.3 BERT + Max Pooling + L2 Normalization with K-Means on Emotion Dataset

This experiment evaluates clustering performance on the Emotion dataset using BERT (bert-base-uncased) embeddings with max pooling and L2 normalization, followed by k-means clustering with six clusters corresponding to the emotion categories.

The obtained results are the following:

- **ACC**: 19.94%
- **NMI**: 0.61%
- **ARI**: 0.33%

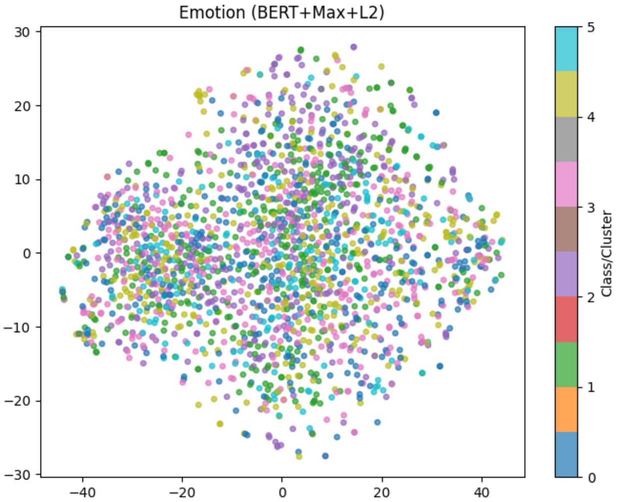

Figure 23: t-SNE visualization of BERT embeddings with max pooling and L2 normalization on the Emotion dataset.

The t-SNE visualization shows a fully collapsed embedding space with no discernible cluster structure. All emotion categories are uniformly mixed, forming a single dense cloud without visible substructures or class-specific regions.

This indicates that, in this configuration, BERT embeddings do not encode emotion-specific separability in a way that is exploitable by distance-based clustering methods. The representation appears to be dominated by non-semantic factors such as syntactic or lexical overlap, rather than affective content.

As a consequence, k-means fails to identify meaningful centroids, resulting in performance close to random assignment across all evaluation metrics.

7 Interpretability Analysis of Best-performing Configurations

To better understand the behavior of the proposed pipelines, the five best-performing configurations across all datasets are analyzed. Interpretability is addressed using a post-hoc analysis approach based on TCRE. A linear classifier is trained to predict cluster assignments from sparse lexical features (e.g., bag-of-words representations), and the learned weights are used to extract *indicative terms* for each cluster. These terms provide human-readable explanations of cluster semantics and allow identification of potential failure cases.

7.1 AG News: ELMo + Mean Pooling + LayerNorm + KMeans

This configuration achieves the best overall performance on AG News. The interpretability analysis is conducted using TCRE, which extracts indicative terms for each cluster.

Cluster 0: space, scientists, afp, software, study, technology, sunday, web, researchers, service

Cluster 1: oil, shares, market, profit, investors, quarter, stocks, july, earnings, sales

Cluster 2: iraq, minister, leader, israel, president, india, police, people, rebels, israeli

Cluster 3: athens, olympics, champion, olympic, league, coach, night, company, team, gold

Cluster 0 clearly corresponds to the *Sci/Tech* category, as shown by the presence of terms related to research and technology. The vocabulary is coherent and domain-specific, indicating a well-formed cluster.

Cluster 1 is strongly associated with the *Business* category. Financial and economic terms dominate the cluster, suggesting that this topic is effectively captured in the embedding space.

Cluster 2 aligns with the *World* category, with a focus on geopolitical entities, conflicts, and international actors. The high density of country and leadership-related terms reflects a clear semantic grouping.

Cluster 3 corresponds to the *Sports* category. Terms related to competitions and teams dominate, although the presence of a generic term such as *company* suggests minor noise within the cluster.

Overall, the clusters are highly interpretable and show a strong alignment with the underlying ground-truth categories, confirming the quality of the learned representation.

7.2 Results on DBpedia (ELMo + Mean + Identity + KMeans)

- Cluster 1: genus, species, family, moth, plant, orchid, native, cultivar, following, described
- Cluster 2: born, la, singer, italian, musician, painter, politician, en, hesse, da
- Cluster 3: film, directed, movie, films, play, actress, comedian, manga, actor, plays
- Cluster 4: village, municipality, district, arabic, province, india, situated, located, ukrainian, england
- Cluster 5: politician, mayor, minister, member, served, elected, born, civil, republican, governor
- Cluster 6: river, lake, mountain, tributary, reservoir, located, romania, hills, school, elevation
- Cluster 7: album, band, music, label, released, records, songwriter, record, rock, best
- Cluster 8: book, novel, published, books, author, fiction, magazine, poet, edition, written
- Cluster 9: church, historic, museum, built, located, restaurant, house, park, hotel, school
- Cluster 10: company, media, products, business, bank, manufacturer, commercial, based, research, journal
- Cluster 11: aircraft, car, built, produced, designed, manufactured, motorcycle, developed, seat, locomotive
- Cluster 12: born, footballer, competed, player, played, league, team, professional, cricketer, olympics

- Cluster 13: navy, ship, class, built, vessel, royal, ships, war, steamer, service

The extracted terms reveal a highly structured clustering aligned with the semantic organization of DBpedia. Most clusters correspond to well-defined ontological categories, such as biological taxonomy (Cluster 1), artistic professions (Cluster 2), films and entertainment (Cluster 3), and geographical entities (Cluster 4).

Some redundancy emerges across clusters sharing similar lexical cues. For instance, the term *born* appears in multiple clusters (e.g., Cluster 2, 5, and 12), reflecting the common structure of biographical entries in DBpedia. Despite this overlap, the surrounding terms provide sufficient context to disambiguate categories (e.g., sports vs politics vs arts).

Clusters related to media and cultural production (Clusters 3, 7, and 8) are particularly coherent, each focusing on a distinct domain (films, music, and books). Similarly, infrastructure and man-made entities are well separated across clusters such as buildings (Cluster 9), companies (Cluster 10), and vehicles (Cluster 11).

Overall, the TCRE explanations confirm that the embedding space preserves high-level semantic groupings, even without normalization. The clustering captures both topical and structural regularities of the dataset, with minor ambiguities mainly due to shared lexical patterns across entity types.

7.3 Results on AG News (ELMo + Mean + ℓ_2 + KMeans)

- Cluster 0: space, scientists, software, afp, study, technology, sunday, web, researchers, security
- Cluster 1: athens, olympics, champion, olympic, coach, league, company, night, gold, team
- Cluster 2: minister, iraq, leader, president, israel, india, police, people, israeli, rebels
- Cluster 3: oil, market, shares, profit, investors, quarter, july, stocks, earnings, sales

The cluster structure remains clearly interpretable, with each group corresponding to a distinct news category. Technology and science-related content is captured by Cluster 0, while sports is well represented in Cluster 1 through terms such as *olympics*, *team*, and *coach*. Cluster 2 groups geopolitical and international news, and Cluster 3 captures financial and business-related topics.

Compared to the LayerNorm configuration, the separation between topics is still evident, but slightly less clean. Some generic terms (e.g., *company*) appear in clusters where they are not central, suggesting a mild loss of specificity. This is consistent with the effect of ℓ_2 normalization, which reduces magnitude differences but may blur distinctions between semantically related documents.

Overall, the explanations confirm that the model preserves high-level topic separability, although with slightly weaker lexical precision.

7.4 Results on DBpedia (ELMo + Mean + ℓ_2 + KMeans)

- Cluster 0: album, band, released, music, label, records, record, rock, singer, songwriter
- Cluster 1: university, school, england, hospital, st, college, located, church, london, secondary
- Cluster 2: born, painter, la, singer, italian, writer, japanese, politician, poet, germany
- Cluster 3: company, media, bank, manufacturer, business, products, born, based, headquartered, independent
- Cluster 4: politician, mayor, member, minister, elected, governor, served, party, assembly, republican
- Cluster 5: aircraft, car, produced, built, designed, manufactured, developed, motorcycle, air, seat
- Cluster 6: genus, species, family, moth, plant, orchid, native, cultivar, described, includes
- Cluster 7: historic, built, park, church, located, hotel, museum, house, building, restaurant
- Cluster 8: village, district, municipality, province, arabic, castle, azerbaijan, azerbaijani, catholic, india
- Cluster 9: book, novel, published, books, author, fiction, magazine, history, written, literature
- Cluster 10: film, directed, movie, films, play, actor, series, comedian, comedy, novel
- Cluster 11: river, lake, mountain, tributary, located, reservoir, romania, hills, hill, elevation
- Cluster 12: navy, ship, built, class, ships, vessel, war, royal, coast, service
- Cluster 13: born, footballer, competed, player, played, team, league, olympics, cricketer, football

The clustering remains highly interpretable and aligned with DBpedia ontology categories. Distinct semantic groups clearly emerge, including music (Cluster 0), education and institutions (Cluster 1), biographies (Cluster 2), companies (Cluster 3), politics (Cluster 4), vehicles (Cluster 5), biological taxonomy (Cluster 6), and geographical entities (Clusters 8 and 11).

Compared to the identity normalization setting, the overall structure is preserved, but with slightly increased mixing across clusters. This is visible in the recurrence of generic terms such as *born* across multiple clusters (e.g., Cluster 2, 3, and 13), indicating reduced lexical sharpness.

Despite this, most clusters remain semantically coherent and easily interpretable. Cultural domains (music, books, films) and technical domains (vehicles, ships, companies) are still well separated, suggesting that ℓ_2 normalization does not significantly disrupt the global organization of the embedding space.

Overall, the TCRE explanations confirm that the representation retains strong semantic grouping, with minor degradation in cluster purity due to normalization effects.

7.5 Results on Yahoo (ELMo + Mean + ℓ_2 + KMeans)

- Cluster 0: love, girlfriend, girls, boyfriend, sex, girl, life, feelings, women, romantic
- Cluster 1: god, religion, jesus, bible, constitution, iraq, believe, christian, muslims, christianity
- Cluster 2: computer, software, email, file, download, yahoo, spam, server, pc, files
- Cluster 3: treatment, doctor, blood, eat, health, causes, body, sleep, caused, disease
- Cluster 4: song, dont, riddle, help, pay, movie, basketball, baseball, christmas, hottest
- Cluster 5: french, nc, el, david, william, africa, capital, 1969, pi, canada
- Cluster 6: credit, pay, school, college, court, police, money, job, cards, passport
- Cluster 7: earth, temperature, carbon, light, water, wind, dna, atoms, inches, volume
- Cluster 8: com, site, http, search, org, website, click, linked, link, id
- Cluster 9: ya, lol, bush, ganguly, dead, im, army, program, uk, guy

The interpretability analysis highlights a much noisier and less structured clustering compared to previous datasets. While some clusters remain clearly interpretable, such as relationships (Cluster 0), religion (Cluster 1), technology (Cluster 2), health (Cluster 3), and science (Cluster 7), several others exhibit mixed or weak semantic coherence.

Cluster 4, for instance, contains heterogeneous terms spanning entertainment, sports, and generic queries, suggesting a lack of a dominant topic. Similarly, Cluster 5 appears highly fragmented, mixing names, locations, and numerical references without a clear unifying theme.

Other clusters reflect functional rather than semantic grouping. Cluster 8 captures web-related tokens (e.g., *http*, *site*, *link*), likely driven by formatting patterns rather than topic similarity. Cluster 9 includes informal and noisy tokens (e.g., *lol*, *ya*, *im*), indicating that stylistic features influence the clustering process.

Overall, the TCRE explanations confirm that, in the Yahoo dataset, the embedding space struggles to produce well-separated semantic clusters. High lexical overlap, informal language, and heterogeneous topics lead to blurred cluster boundaries and reduced interpretability, consistent with the lower quantitative performance observed.

7.6 Results on R5 (ELMo + Mean + LN + KMeans)

- Cluster 0: dollar, bank, fed, dealers, yen, bundesbank, treasury, market, lt, deficit
- Cluster 1: vs, 1987, said, says, cts, shr, 19, qtr, mar, stg
- Cluster 2: trade, lt, vs, oil, said, acquire, takeover, group, crude, dlrs
- Cluster 3: oil, crude, barrel, barrels, petroleum, drilling, opec, pipeline, light, gasoline
- Cluster 4: trade, reagan, says, nations, louvre, goods, baker, miyazawa, minister, stoltenberg

The clusters exhibit a relatively clear thematic structure, reflecting the domain-specific nature of the Reuters R5 dataset. Financial topics are strongly represented in Cluster 0, with terms related to currency, banking, and monetary policy. Cluster 3 is highly coherent and focused on the oil sector, with specialized terminology such as *opec*, *barrel*, and *pipeline*.

Clusters 2 and 4 capture broader economic and trade-related content. Cluster 2 emphasizes corporate activities such as acquisitions and market operations, while Cluster 4 includes geopolitical and policy-related terms associated with international trade.

Cluster 1 appears less interpretable, dominated by abbreviations and numerical tokens (e.g., *cts*, *shr*, *qtr*), which are typical of financial reports but do not define a clear standalone topic. This suggests that some clusters are influenced more by formatting conventions than by semantic content.

Overall, the TCRE explanations indicate that the embedding space captures domain-specific structure effectively, particularly for well-defined sectors such as finance and energy. However, some overlap between economic subtopics and the presence of generic reporting tokens slightly reduce cluster purity.

8 Conclusion

This work has investigated unsupervised document clustering based on deep pretrained language model representations, with a focus on ELMo and BERT embeddings. The analysis has systematically examined the impact of pooling strategies, normalization techniques, and clustering algorithms across multiple benchmark datasets.

The experimental results highlight that the structure of the embedding space plays a central role in determining clustering quality. In particular, datasets with strong topical coherence, such as Reuters R5 and AG News, tend to produce well-separated clusters, especially when mean pooling is combined with appropriate normalization techniques such as layer normalization or ℓ_2 normalization. In these settings, both k-means and agglomerative clustering are able to recover meaningful groupings, with only minor differences between the two approaches.

In contrast, datasets characterized by higher semantic overlap and informal language, such as Yahoo Answers and the Emotion dataset, exhibit severely entangled embedding spaces. In these cases, clustering performance drops significantly regardless of the model or clustering algorithm used, indicating that the representational capacity of the considered embeddings is insufficient to induce clear separability in low-context or highly ambiguous settings.

Across all experiments, normalization is observed to be a critical factor in shaping the geometry of the embedding space. While layer normalization tends to produce more stable and compact clusters, ℓ_2 normalization shows dataset-dependent behavior, sometimes improving and sometimes degrading separability. Pooling strategies also play a significant role, with mean pooling generally providing more consistent performance compared to max pooling in the evaluated settings.

Finally, the comparison between k-means and agglomerative clustering shows that no single clustering method is universally superior. Instead, performance differences are largely driven by the structure of the learned representation space, with centroid-based methods performing better in more isotropic settings and hierarchical methods offering advantages in cases with local substructure.

The interpretability analysis based on TCRE further supports these findings, showing that well-performing configurations correspond to semantically coherent and human-interpretable clusters, while poorly performing settings produce noisy and less meaningful term distributions. This confirms that cluster quality is tightly linked not only to quantitative metrics but also to the underlying semantic organization of the embedding space.